

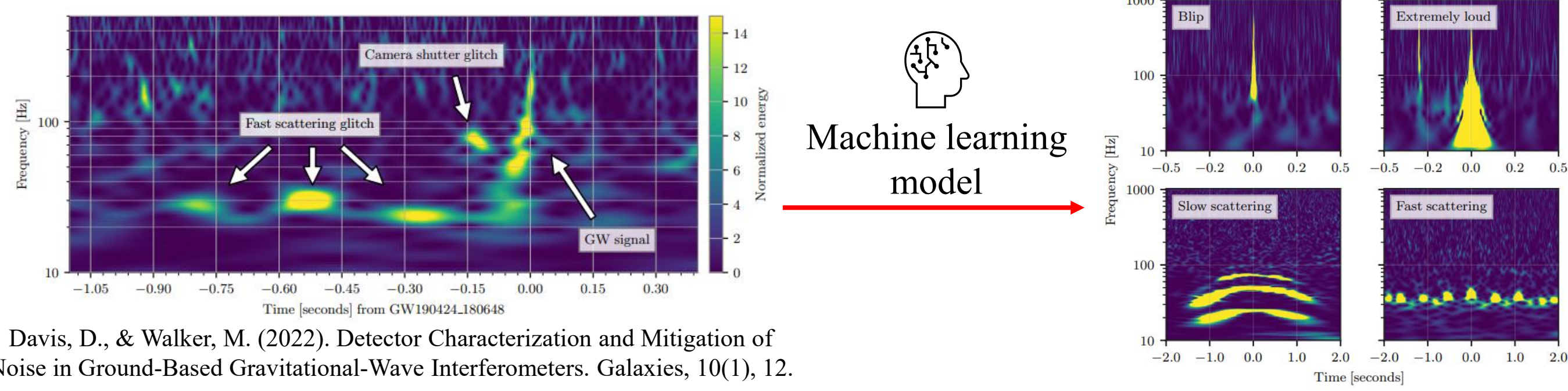


Abstract

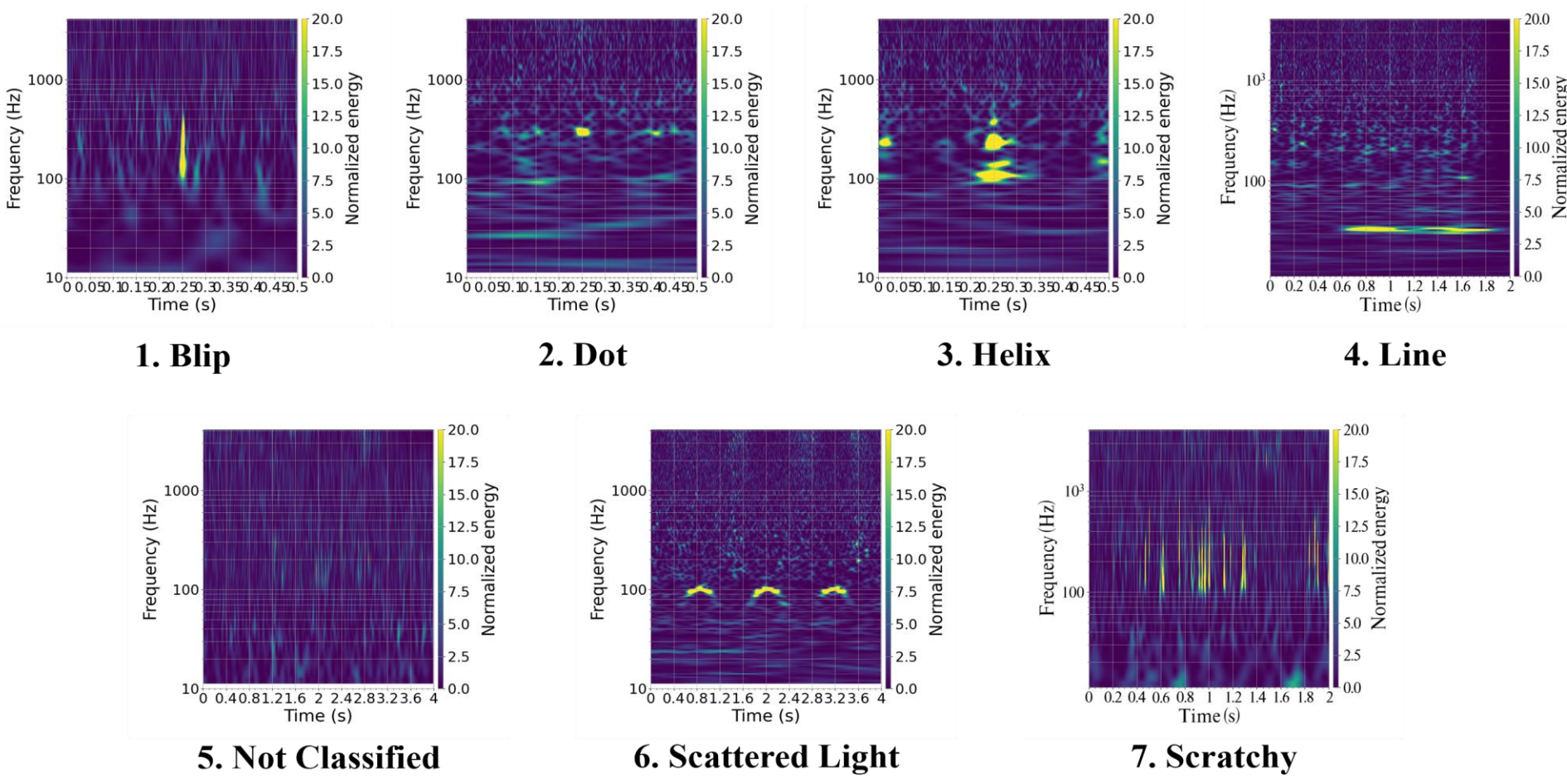
Ground-based gravitational wave detectors are facing increasing transient noise, i.e., glitches, that hinder signal detection and limit detector sensitivity. While classifying these glitches is crucial for data quality, conventional machine learning approaches depend on labor-intensive data labeling, which is unsustainable for evolving noise types. To address this challenge, we investigate an unsupervised deep learning method that is capable of discovering patterns without labeled data. Specifically, we employ deep embedded clustering, an unsupervised method that jointly optimizes feature representation and cluster assignment by utilizing deep neural networks. We apply this method to glitch spectrogram data obtained from the KAGRA detector's observation run. Our results demonstrate that deep embedded clustering effectively identifies cluster structures in KAGRA glitch data. This method offers a robust and scalable solution for future gravitational wave observations.

Introduction

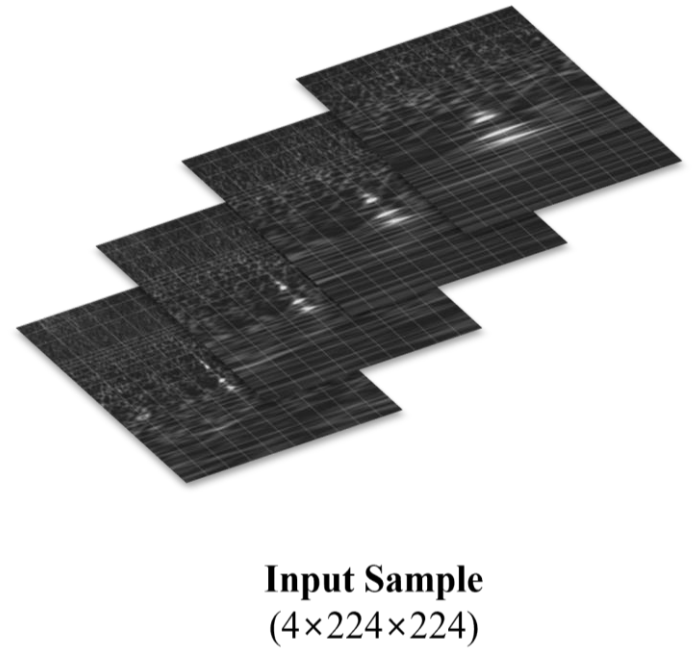
- Since the sensitivity of ground-based gravitational wave detectors has been upgraded, thousands of events and transient noises (glitches) have been recorded.
- Classifying and removing these glitches is critical, as they can mimic or mask real signals. To classify these glitches, especially in large unlabeled datasets, unsupervised learning approaches are essential.
- We apply the Deep Embedded Clustering (DEC) algorithm to improve unsupervised glitch clustering, thereby enhancing data quality for gravitational-wave analysis.



KAGRA O3GK glitch data



Class	Train	Val	Test
Blip	822	105	239
Dot	29	13	5
Helix	197	28	60
Line	304	48	83
Not Classified	65	13	11
Scattered Light	32	5	16
Scratchy	102	9	29
Total	1551	221	443



- Our study uses 8,860 spectrogram images from KAGRA O3GK run (4.7.2020-4.21.2020), spanning 7 glitch types and four timescales: 0.5s; 1s; 2s; 4s.
- Each image visualizes time–frequency features, enabling classification of transient noise.
- For each glitch event, four spectrogram RGB images corresponding to different timescales were cropped and resized into square grayscale images. These four images were then stacked to form a single input sample with a shape of $4 \times 224 \times 224$.

Method

Algorithms

- Deep Embedded Clustering (DEC):** Unsupervised deep learning algorithm that simultaneously learns feature representations and cluster assignments. It uses an autoencoder to compress the data into low dimension and then optimizes the clustering process within that latent space (illustrated on the bottom figure).
- Principal Component Analysis (PCA):** Traditional machine learning algorithm that reduces dimensionality linearly.
- k-means :** Clustering algorithm that partitions data into k clusters by iteratively assigning points to the nearest centroid and updating centroids to minimize within-cluster variance.
- t-SNE :** Nonlinear manifold learning technique that projects high-dimensional data into two or three dimensions for visualization.

Evaluating performance

- Silhouette score :** measuring clustering quality

$$a(i) = \frac{1}{|C_I| - 1} \sum_{j \in C_I, j \neq i} d(i, j) \quad b(i) = \min_{J \neq I} \frac{1}{|C_J|} \sum_{j \in C_J} d(i, j)$$

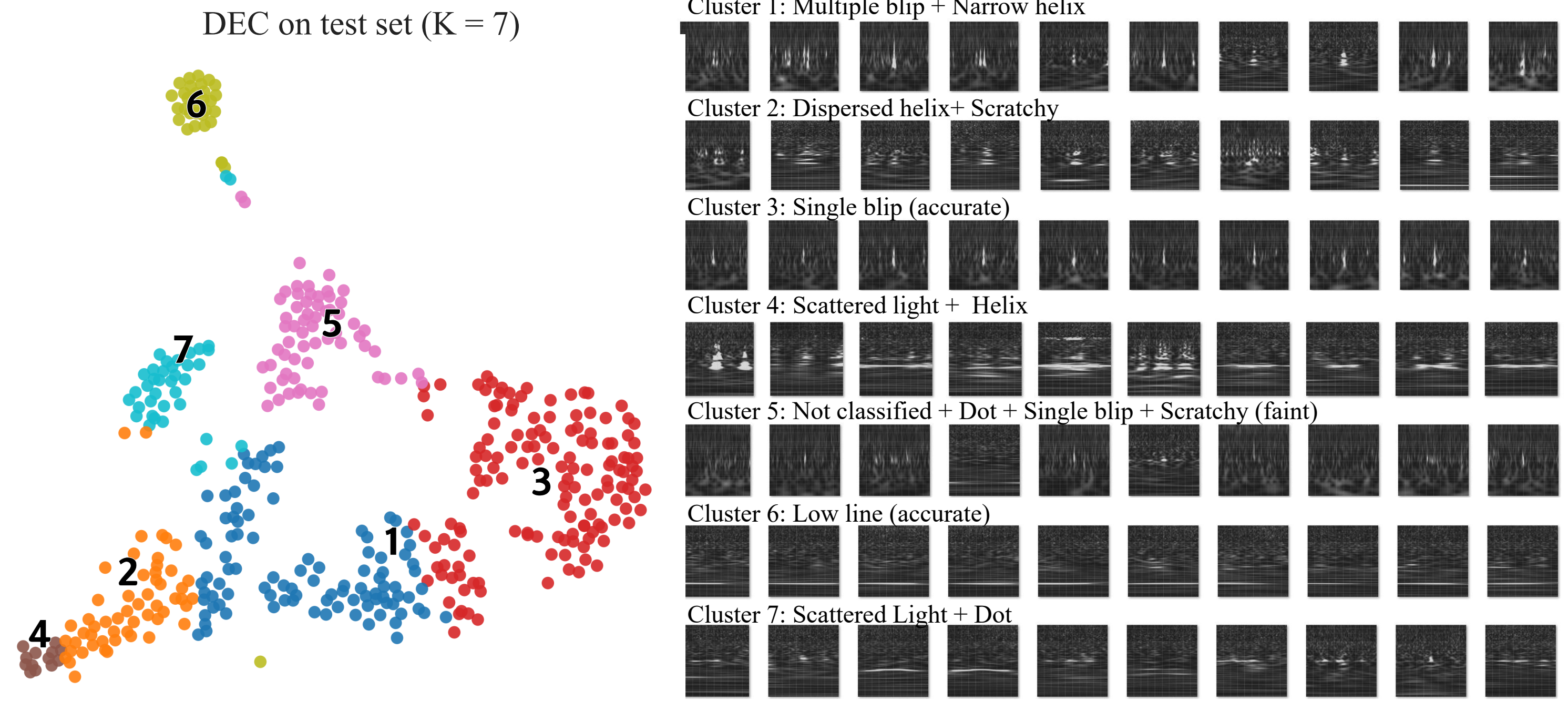
$$s(i) = \begin{cases} \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} & \text{if } |C_I| > 1 \\ 0 & \text{if } |C_I| = 1 \end{cases}, \quad (-1 \leq s(i) \leq 1)$$

- Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) :** NMI indicates the overlap of information between your discovered groups and the true categories; ARI indicates how often the clusters match with the true labels on separating or grouping data points.

$$NMI = \frac{I(Y; \hat{Y})}{\sqrt{H(Y) \times H(\hat{Y})}}, \quad ARI = \frac{RI - E(RI)}{\max(RI) - E(RI)} \quad RI = \frac{C_1 + C_2}{\binom{n}{2}}$$

Xie, J., Girshick, R., & Farhadi, A. (2016). Unsupervised deep embedding for clustering analysis. In *International conference on machine learning* (pp. 478–487). PMLR.

t-SNE visualization of Deep Embedded Clustering algorithms

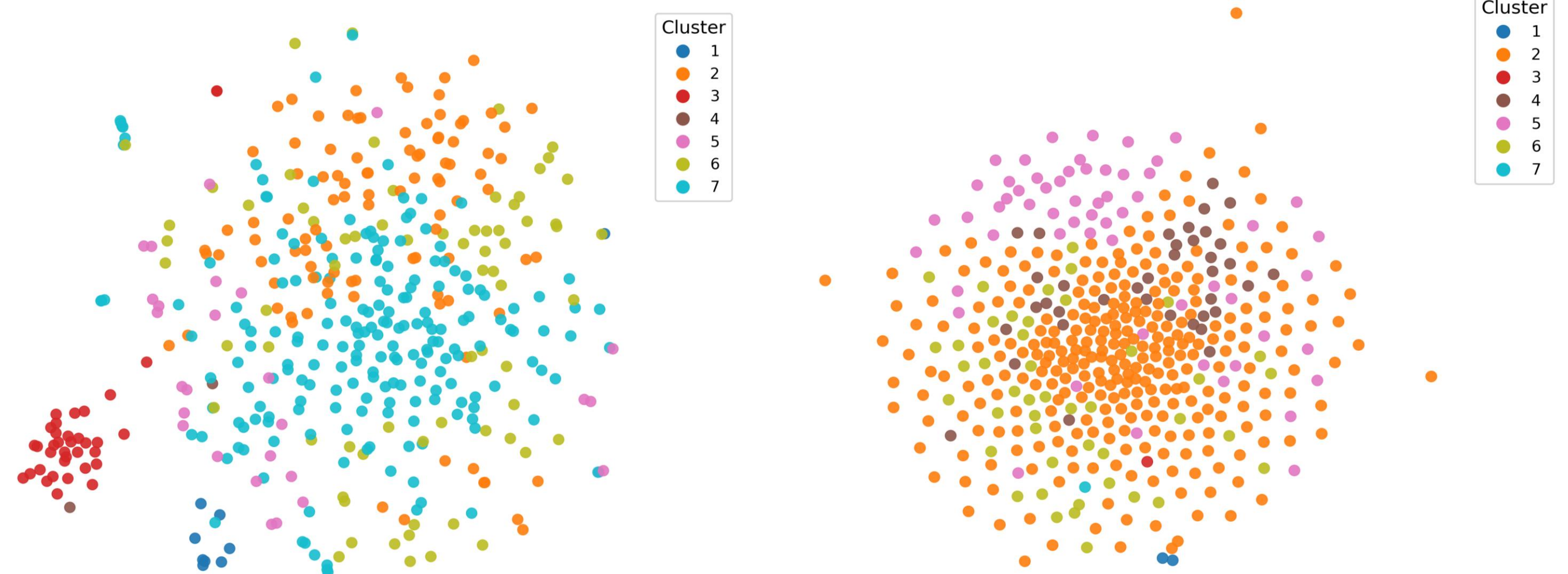


- We trained the DEC algorithm using preprocessed spectrogram image data and applied the model to the test dataset to evaluate clustering performance.
- As shown in the sample distribution of the test set, the t-SNE projection of DEC (left) produces tighter and more distinct clusters compared to other machine learning algorithms (shown in the lower panel).
- The 10 randomly selected glitch samples classified by the DEC algorithm (right) show morphological similarity within each cluster. Glitches belonging to the Blip and Line classes are well grouped together, indicating strong intra-cluster consistency.

Comparison with other machine learning algorithms

PCA + k-means on test set (K=7)

Raw features + k-means on test set (K=7)



- This figure compares the k-means clustering results of seven glitch classes using PCA-based dimensionality reduction (left) and raw data (right).
- Without PCA preprocessing, these classes appear highly entangled and difficult to distinguish clusters.
- With PCA preprocessing, the clusters become more separable but still lack clearly recognizable cluster structures when compared to the DEC method.
- This highlights the effectiveness of DEC in resolving vague and overlapping structures in the KAGRA glitch.

Comparison with LIGO data (Gravity Spy v1.1.0)

Method	DEC	PCA + k-means	Raw feature + k-means
t-SNE visualization on test set data (K=22)			
Clustering performance	ARI : 0.3108 NMI : 0.5761	ARI : 0.2230 NMI : 0.5750	ARI : 0.1942 NMI : 0.5483

- The t-SNE visualization plot on the table shows the result of applying the same method to the LIGO O1 and O2 glitch dataset provided by Gravity Spy, which consists of 7,967 images across 22 classes.
- DEC achieved improved clustering performance compared to PCA and raw features, consistent with the results obtained on the KAGRA O3GK dataset.
- Unlike the KAGRA dataset, the Gravity Spy dataset shows relatively clear cluster structures even when using raw data, with no significant performance difference observed when compared to PCA preprocessing. This indicates that KAGRA glitch distributions exhibit more ambiguous and visually indistinguishable features than LIGO glitch distributions, highlighting the necessity of the DEC method.

Clustering performance evaluation

Method	Silhouette score	NMI	ARI
DEC	0.4347	0.3896	0.3343
PCA+ k-means	0.0196	0.2149	0.1285
Raw Features + k-means	0.0096	0.1736	0.2219

Conclusion

- We demonstrated that the DEC algorithm significantly improves clustering performance on the KAGRA O3GK dataset compared to traditional machine learning methods, enhancing clustering quality (Silhouette score: 0.4347) and accuracy (NMI: 0.3896; ARI: 0.3343).
- These results highlight the effectiveness of deep learning–based approaches in unsupervised learning pipelines, enabling label-free analysis for future detectors and previously unseen glitch types.
- Future work will focus on applying the DEC algorithm to O4 data to identify the cluster structure of KAGRA glitches.